16 – Confidence Intervals

Reference: [ES] 6.1, 6.3

(For pop. mean)

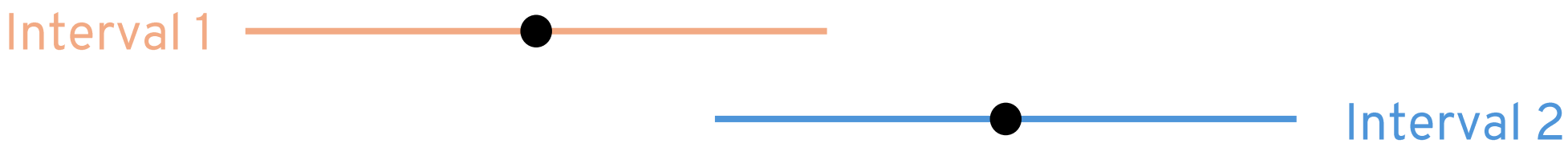
Reference: [ES] 6.1

Fun fact:

Two intervals on the number line have the same length. Then:

1st interval contains the midpoint of 2nd interval

↔ 2nd interval contains the midpoint of 1st interval



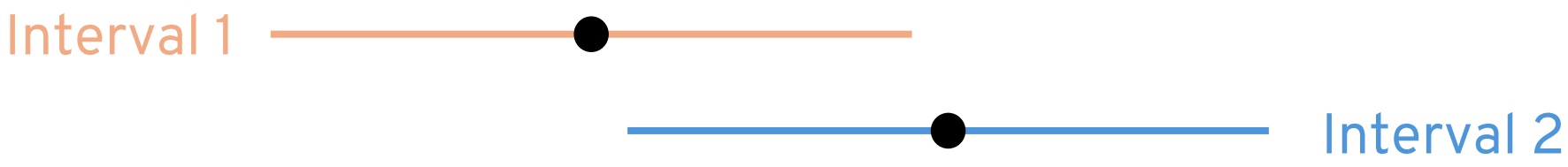
Centers not overlap

Fun fact:

Two intervals on the number line have the same length. Then:

1st interval contains the midpoint of 2nd interval

↔ 2nd interval contains the midpoint of 1st interval



Centers not overlap

Fun fact:

Two intervals on the number line have the same length. Then:

1st interval contains the midpoint of 2nd interval

↔ 2nd interval contains the midpoint of 1st interval

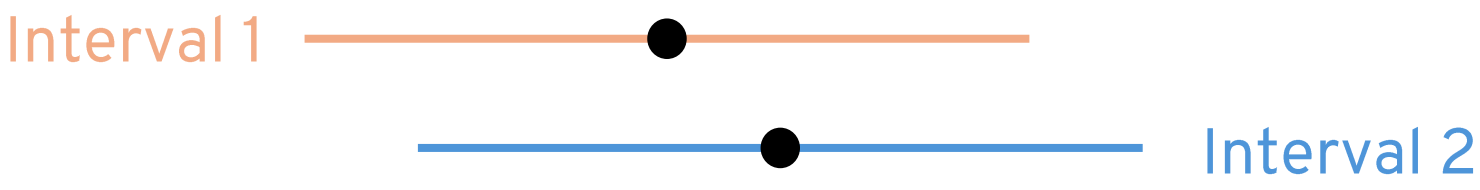
Centers overlap

Fun fact:

Two intervals on the number line have the same length. Then:

1st interval contains the midpoint of 2nd interval

↔ 2nd interval contains the midpoint of 1st interval



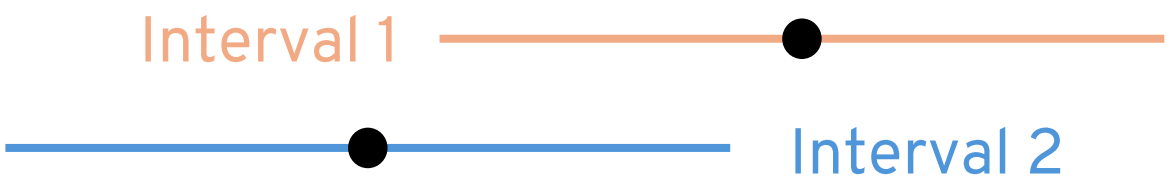
Centers overlap

Fun fact:

Two intervals on the number line have the same length. Then:

1st interval contains the midpoint of 2nd interval

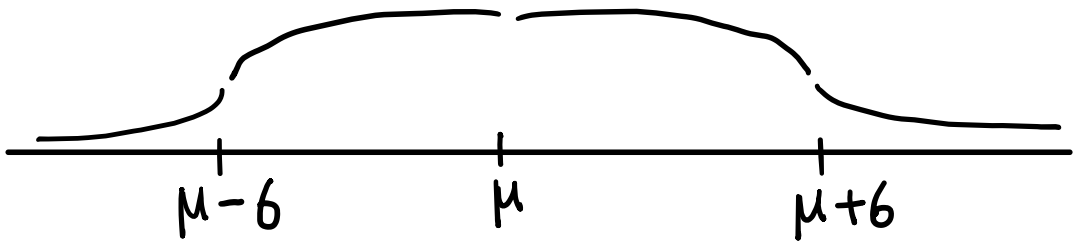
↔ 2nd interval contains the midpoint of 1st interval



Centers not overlap

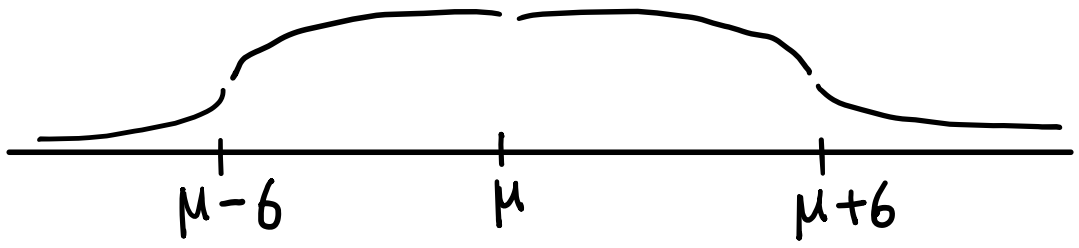
Confidence Interval (Intuition)

Population is normally distributed with $\mu = ?$ and $\sigma = 6$:



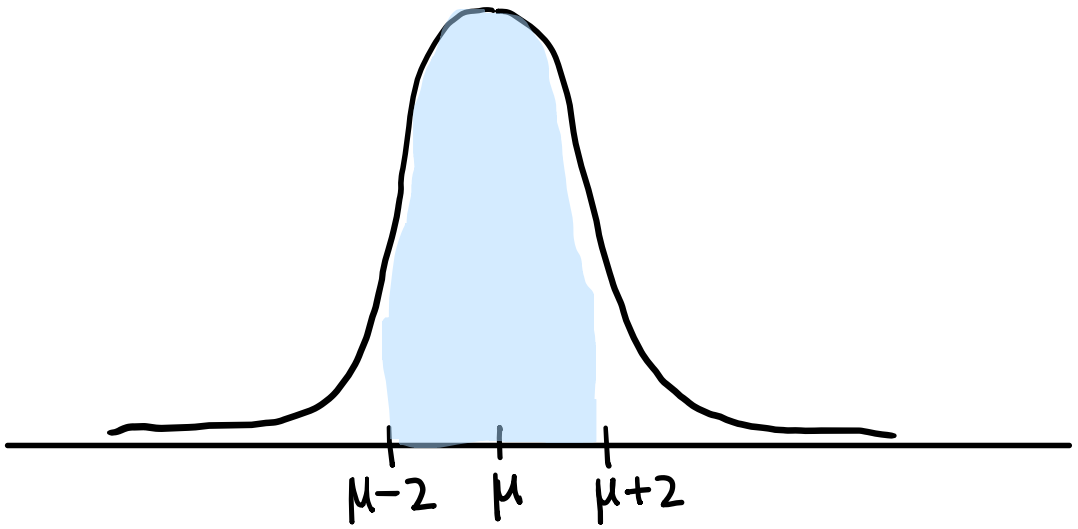
Confidence Interval (Intuition)

Population is normally distributed with $\mu = ?$ and $\sigma = 6$:



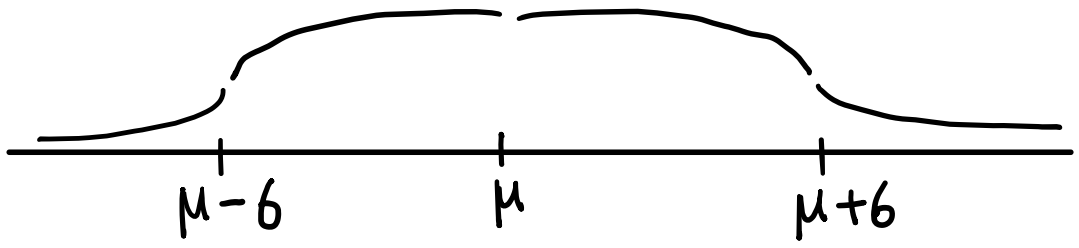
Samples of size $n = 9$ have sample mean $\bar{x}$ distributed like $\mu_{\bar{x}} = \mu$ and $\sigma_{\bar{x}} = \frac{6}{\sqrt{9}} = 2$:

$\bar{x}$ lands in interval $\mu \pm 1 \cdot \sigma_{\bar{x}}$ about 68% of the time of all resamplings.



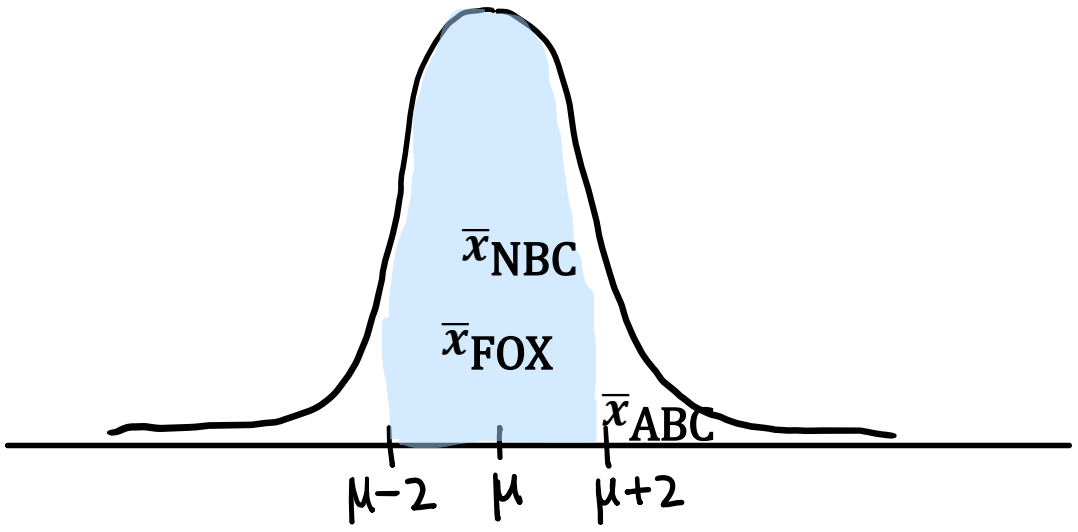
Confidence Interval (Intuition)

Population is normally distributed with $\mu = ?$ and $\sigma = 6$:



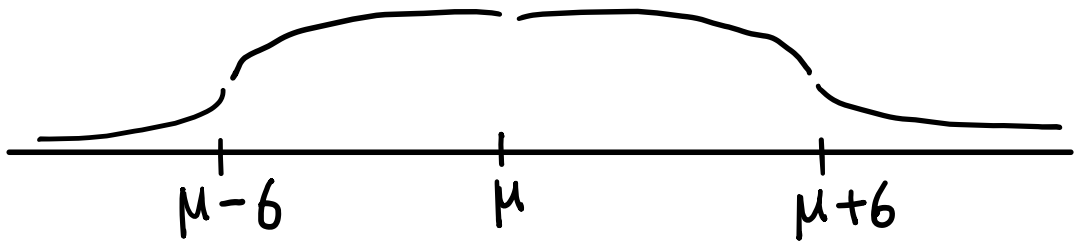
Samples of size $n = 9$ have sample mean $\bar{x}$ distributed like $\mu_{\bar{x}} = \mu$ and $\sigma_{\bar{x}} = \frac{6}{\sqrt{9}} = 2$:

$\bar{x}$ lands in interval $\mu \pm 1 \cdot \sigma_{\bar{x}}$ about 68% of the time of all resamplings.



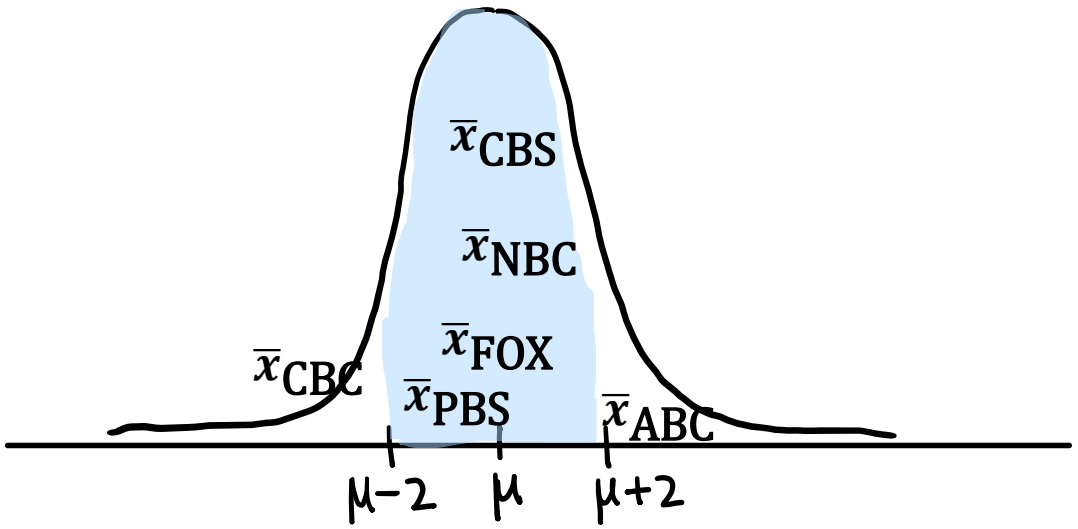
Confidence Interval (Intuition)

Population is normally distributed with $\mu = ?$ and $\sigma = 6$:



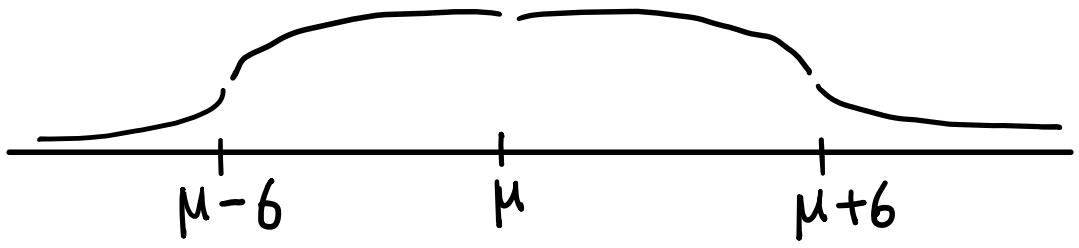
Samples of size $n = 9$ have sample mean $\bar{x}$ distributed like $\mu_{\bar{x}} = \mu$ and $\sigma_{\bar{x}} = \frac{6}{\sqrt{9}} = 2$:

$\bar{x}$ lands in interval $\mu \pm 1 \cdot \sigma_{\bar{x}}$ about 68% of the time of all resamplings.

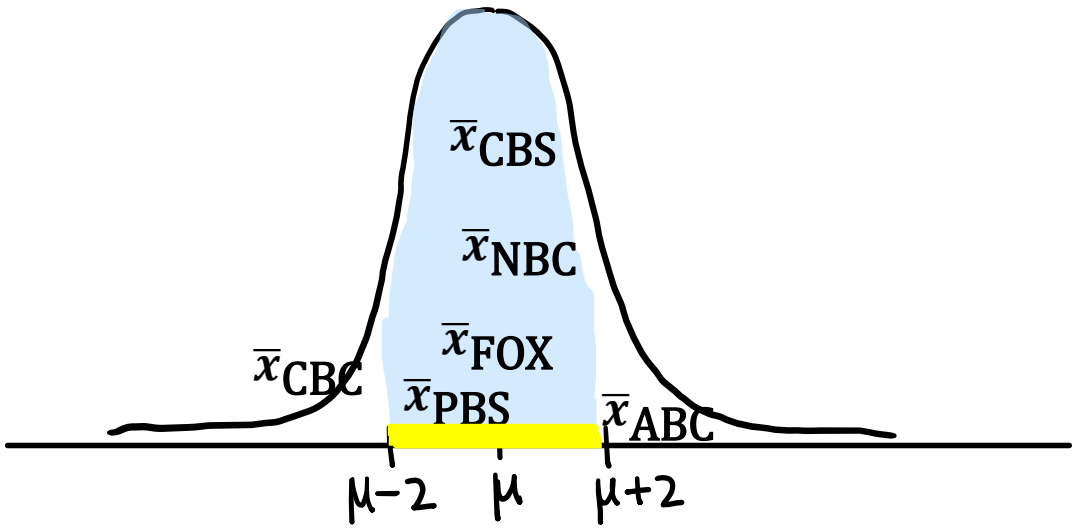


Confidence Interval (Intuition)

Population is normally distributed with $\mu = ?$ and $\sigma = 6$:



Samples of size $n = 9$ have sample mean $\bar{x}$ distributed like $\mu_{\bar{x}} = \mu$ and $\sigma_{\bar{x}} = \frac{6}{\sqrt{9}} = 2$:

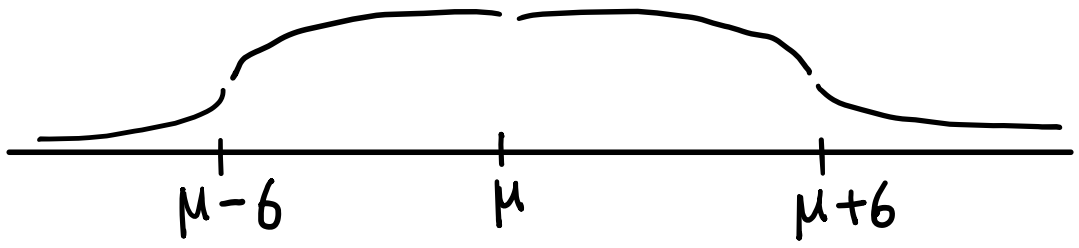


$\bar{x}$ lands in interval $\mu \pm 1 \cdot \sigma_{\bar{x}}$ about 68% of the time of all resamplings.

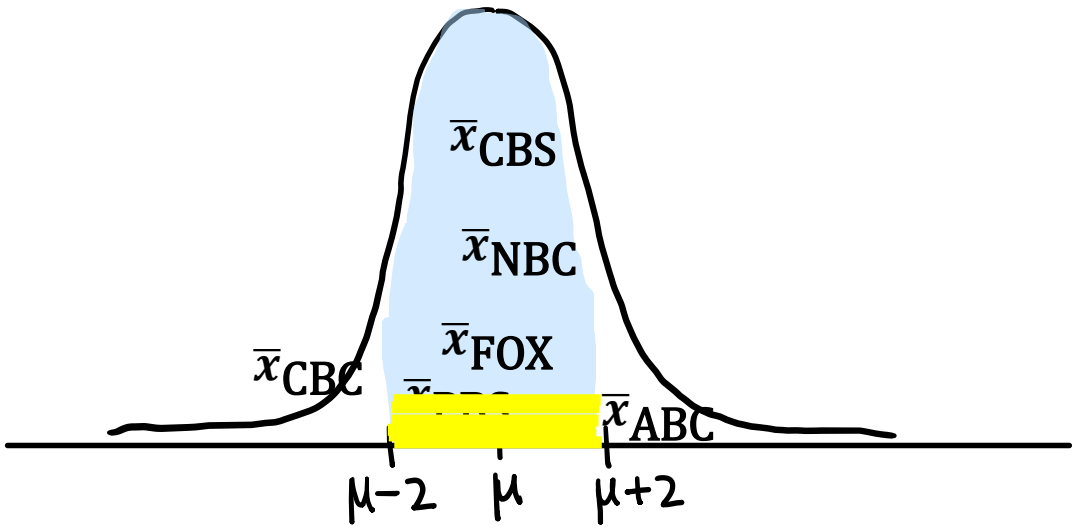
Flipped version: μ lands in interval $\bar{x} \pm 1 \cdot \sigma_{\bar{x}}$ 68% of the time of all resamplings.

Confidence Interval (Intuition)

Population is normally distributed with $\mu = ?$ and $\sigma = 6$:



Samples of size $n = 9$ have sample mean $\bar{x}$ distributed like $\mu_{\bar{x}} = \mu$ and $\sigma_{\bar{x}} = \frac{6}{\sqrt{9}} = 2$:

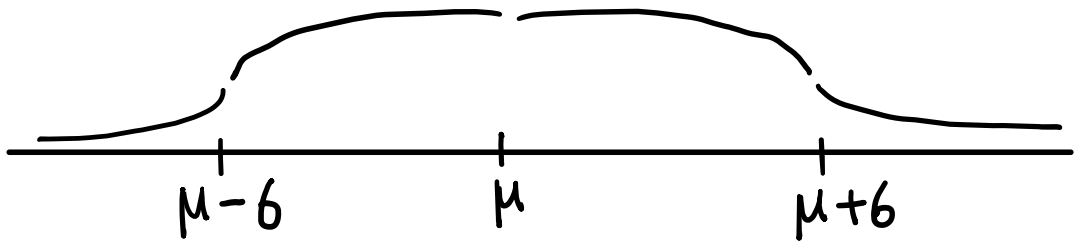


$\bar{x}$ lands in interval $\mu \pm 1 \cdot \sigma_{\bar{x}}$ about 68% of the time of all resamplings.

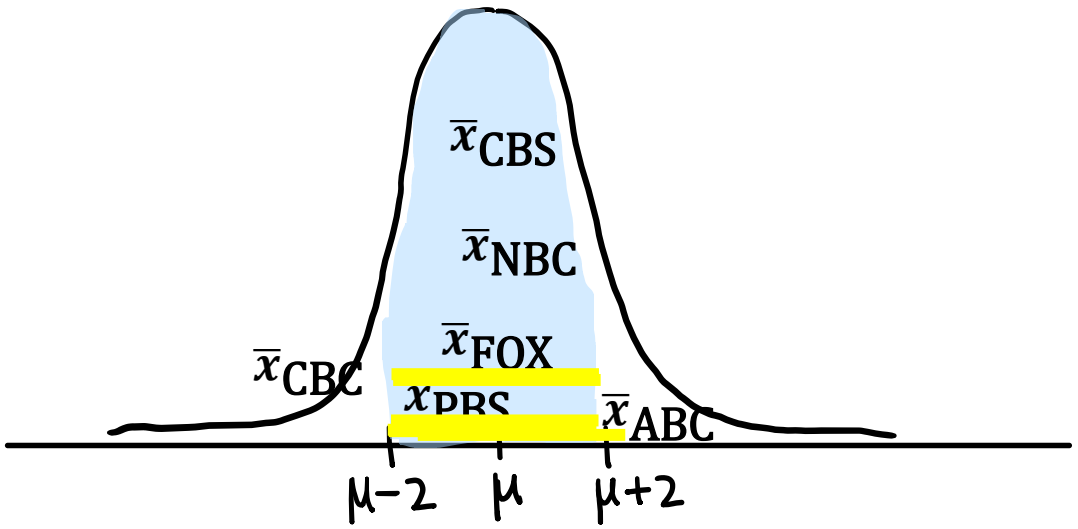
Flipped version: μ lands in interval $\bar{x} \pm 1 \cdot \sigma_{\bar{x}}$ 68% of the time of all resamplings.

Confidence Interval (Intuition)

Population is normally distributed with $\mu = ?$ and $\sigma = 6$:



Samples of size $n = 9$ have sample mean $\bar{x}$ distributed like $\mu_{\bar{x}} = \mu$ and $\sigma_{\bar{x}} = \frac{6}{\sqrt{9}} = 2$:

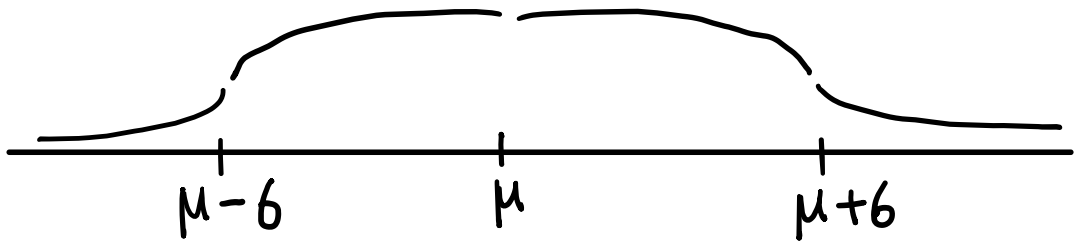


$\bar{x}$ lands in interval $\mu \pm 1 \cdot \sigma_{\bar{x}}$ about 68% of the time of all resamplings.

Flipped version: μ lands in interval $\bar{x} \pm 1 \cdot \sigma_{\bar{x}}$ 68% of the time of all resamplings.

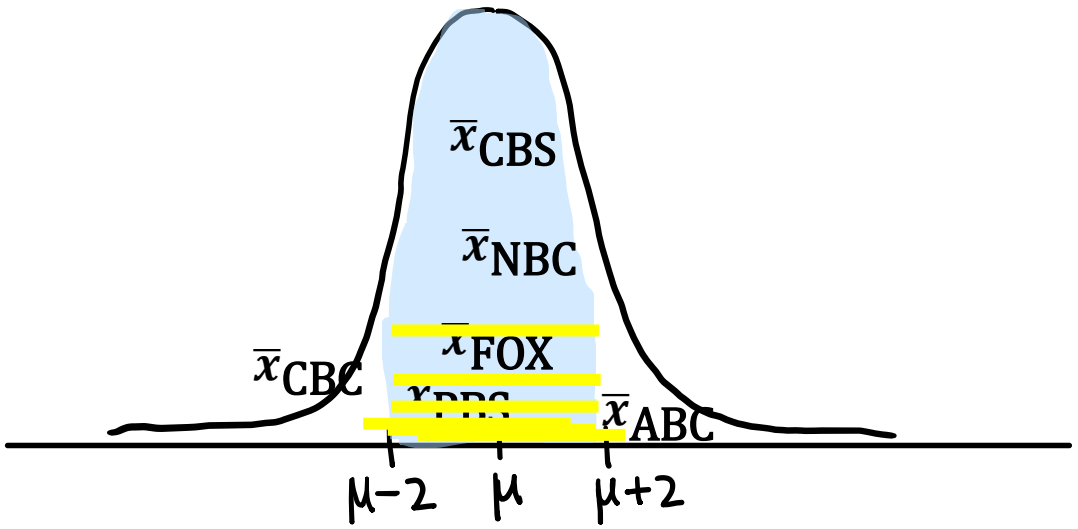
Confidence Interval (Intuition)

Population is normally distributed with $\mu = ?$ and $\sigma = 6$:



Samples of size $n = 9$ have sample mean $\bar{x}$ distributed like $\mu_{\bar{x}} = \mu$ and $\sigma_{\bar{x}} = \frac{6}{\sqrt{9}} = 2$:

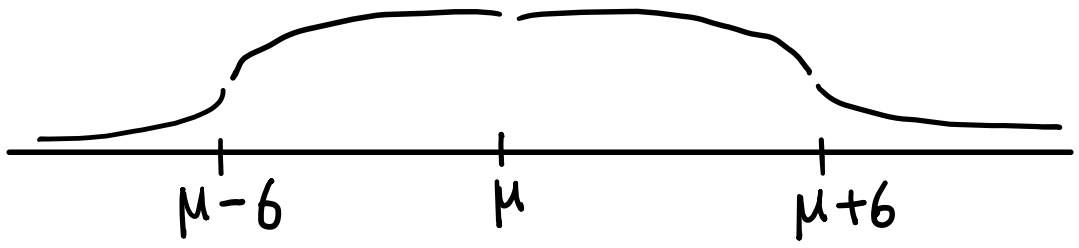
$\bar{x}$ lands in interval $\mu \pm 1 \cdot \sigma_{\bar{x}}$ about 68% of the time of all resamplings.



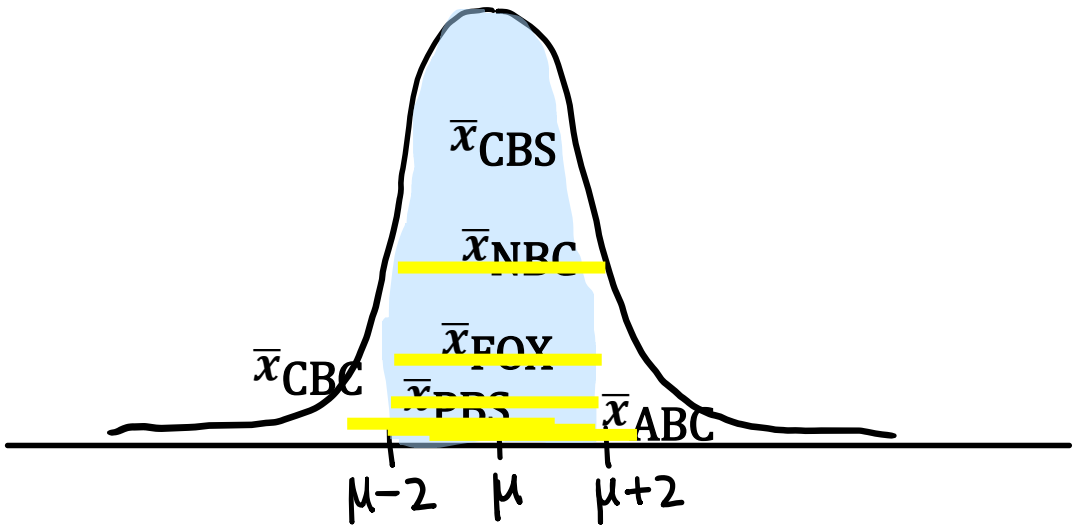
Flipped version: μ lands in interval $\bar{x} \pm 1 \cdot \sigma_{\bar{x}}$ 68% of the time of all resamplings.

Confidence Interval (Intuition)

Population is normally distributed with $\mu = ?$ and $\sigma = 6$:



Samples of size $n = 9$ have sample mean $\bar{x}$ distributed like $\mu_{\bar{x}} = \mu$ and $\sigma_{\bar{x}} = \frac{6}{\sqrt{9}} = 2$:

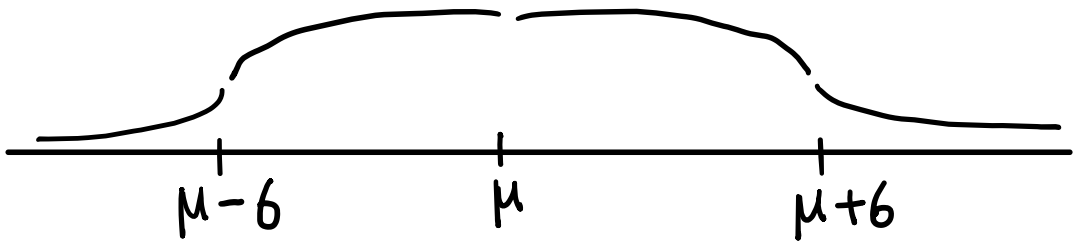


$\bar{x}$ lands in interval $\mu \pm 1 \cdot \sigma_{\bar{x}}$ about 68% of the time of all resamplings.

Flipped version: μ lands in interval $\bar{x} \pm 1 \cdot \sigma_{\bar{x}}$ 68% of the time of all resamplings.

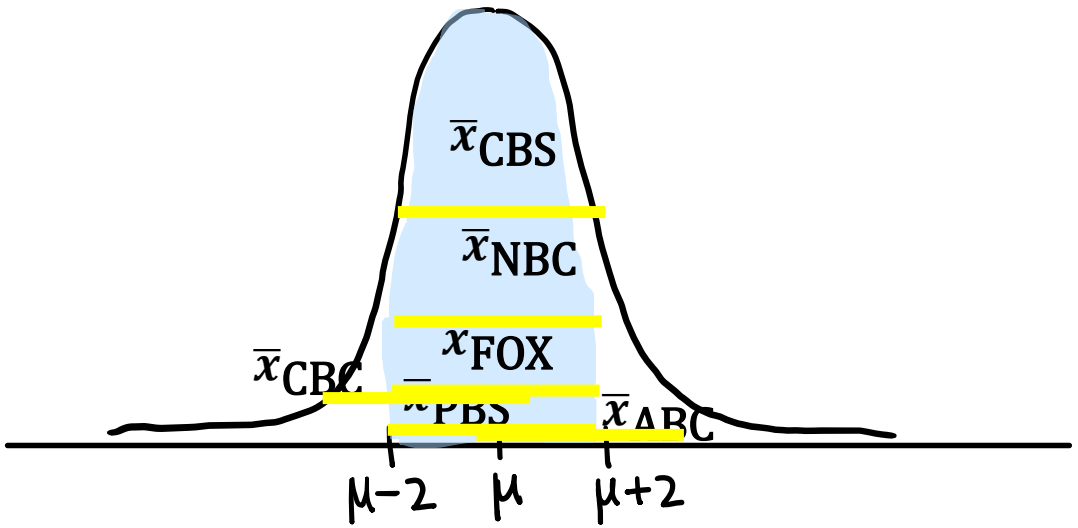
Confidence Interval (Intuition)

Population is normally distributed with $\mu = ?$ and $\sigma = 6$:



Samples of size $n = 9$ have sample mean $\bar{x}$ distributed like $\mu_{\bar{x}} = \mu$ and $\sigma_{\bar{x}} = \frac{6}{\sqrt{9}} = 2$:

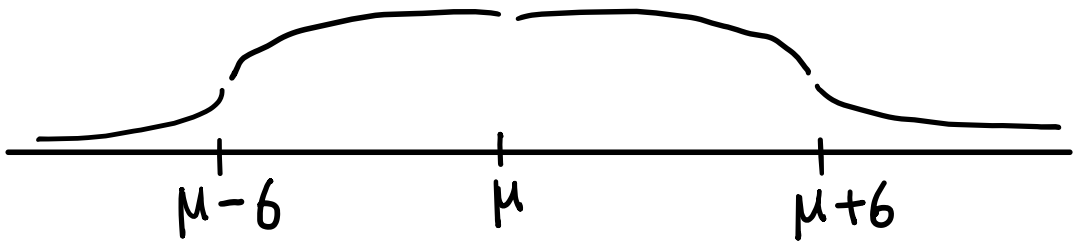
$\bar{x}$ lands in interval $\mu \pm 1 \cdot \sigma_{\bar{x}}$ about 68% of the time of all resamplings.



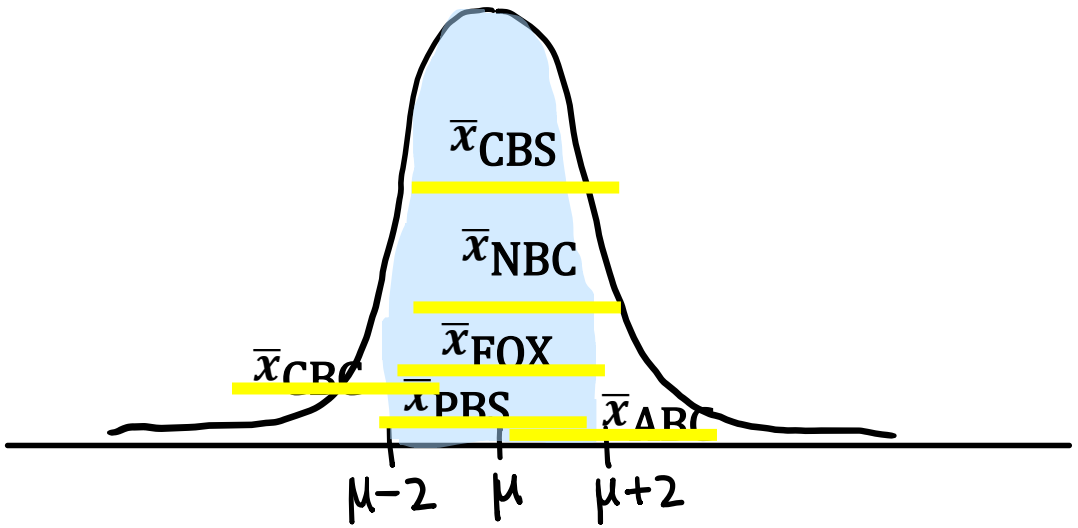
Flipped version: μ lands in interval $\bar{x} \pm 1 \cdot \sigma_{\bar{x}}$ 68% of the time of all resamplings.

Confidence Interval (Intuition)

Population is normally distributed with $\mu = ?$ and $\sigma = 6$:



Samples of size $n = 9$ have sample mean $\bar{x}$ distributed like $\mu_{\bar{x}} = \mu$ and $\sigma_{\bar{x}} = \frac{6}{\sqrt{9}} = 2$:

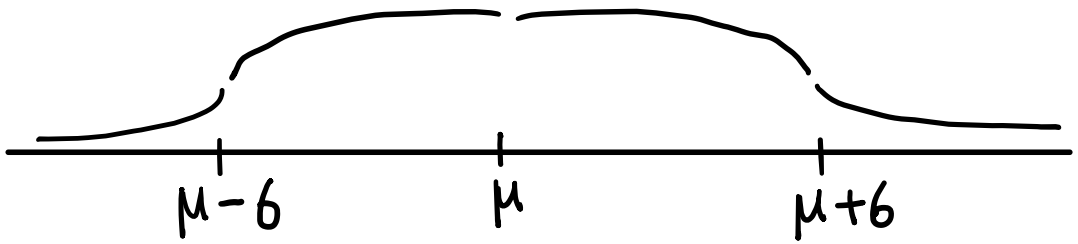


$\bar{x}$ lands in interval $\mu \pm 1 \cdot \sigma_{\bar{x}}$ about 68% of the time of all resamplings.

Flipped version: μ lands in interval $\bar{x} \pm 1 \cdot \sigma_{\bar{x}}$ 68% of the time of all resamplings.

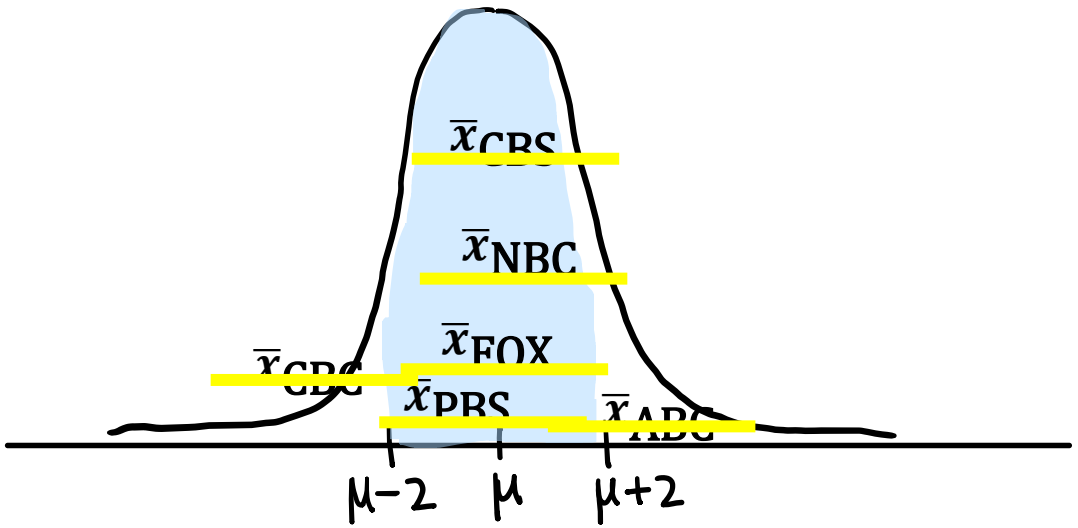
Confidence Interval (Intuition)

Population is normally distributed with $\mu = ?$ and $\sigma = 6$:



Samples of size $n = 9$ have sample mean $\bar{x}$ distributed like $\mu_{\bar{x}} = \mu$ and $\sigma_{\bar{x}} = \frac{6}{\sqrt{9}} = 2$:

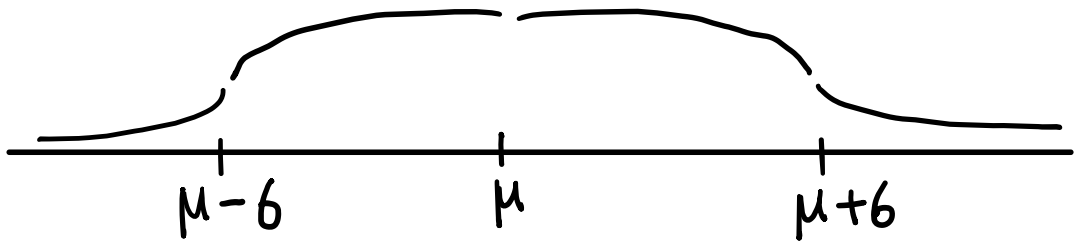
$\bar{x}$ lands in interval $\mu \pm 1 \cdot \sigma_{\bar{x}}$ about 68% of the time of all resamplings.



Flipped version: μ lands in interval $\bar{x} \pm 1 \cdot \sigma_{\bar{x}}$ 68% of the time of all resamplings.

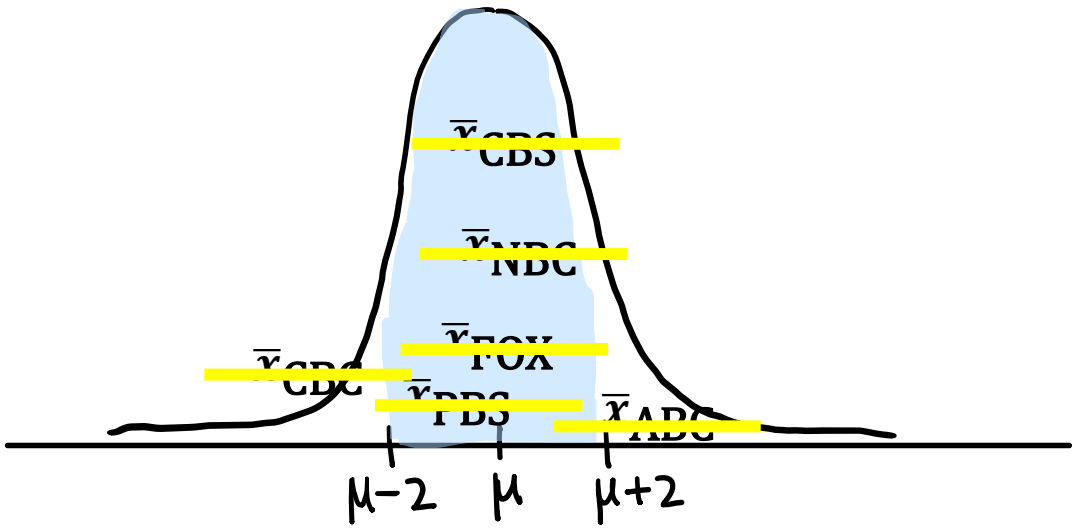
Confidence Interval (Intuition)

Population is normally distributed with $\mu = ?$ and $\sigma = 6$:



Samples of size $n = 9$ have sample mean $\bar{x}$ distributed like $\mu_{\bar{x}} = \mu$ and $\sigma_{\bar{x}} = \frac{6}{\sqrt{9}} = 2$:

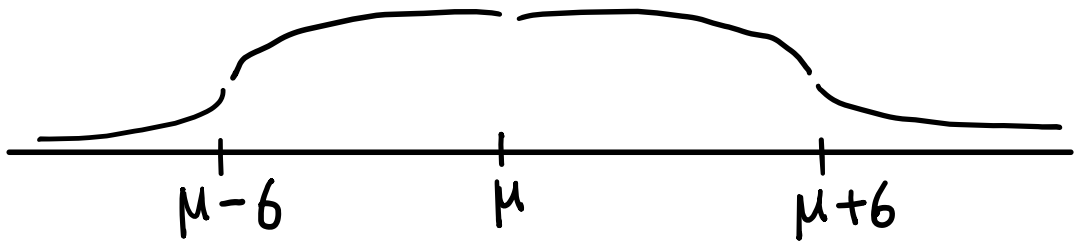
$\bar{x}$ lands in interval $\mu \pm 1 \cdot \sigma_{\bar{x}}$ about 68% of the time of all resamplings.



Flipped version: μ lands in interval $\bar{x} \pm 1 \cdot \sigma_{\bar{x}}$ 68% of the time of all resamplings.

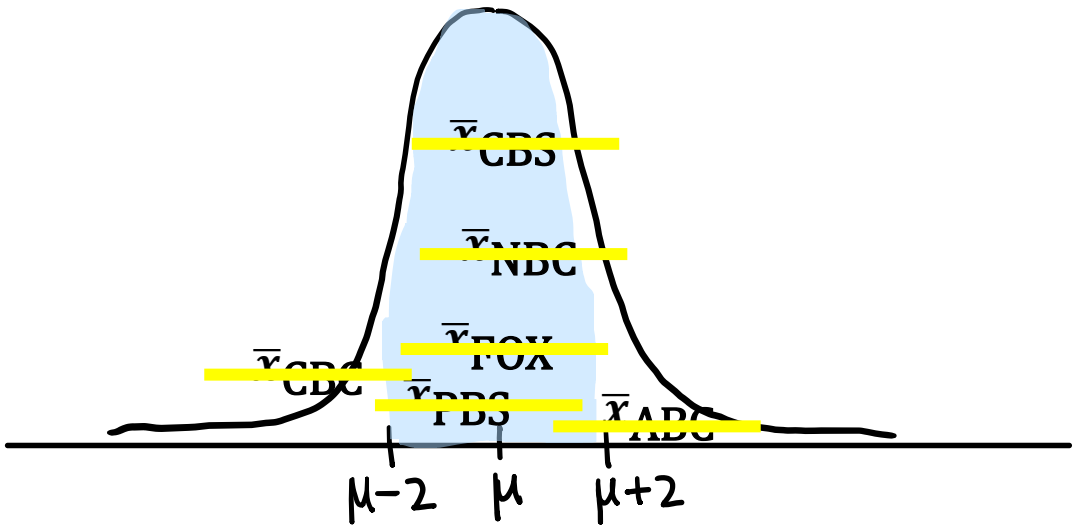
Confidence Interval (Intuition)

Population is normally distributed with $\mu = ?$ and $\sigma = 6$:



Samples of size $n = 9$ have sample mean $\bar{x}$ distributed like $\mu_{\bar{x}} = \mu$ and $\sigma_{\bar{x}} = \frac{6}{\sqrt{9}} = 2$:

$\bar{x}$ lands in interval $\mu \pm 1 \cdot \sigma_{\bar{x}}$ about 68% of the time of all resamplings.



Flipped version: μ lands in interval $\bar{x} \pm 1 \cdot \sigma_{\bar{x}}$ 68% of the time of all resamplings.

"We are 68% confident that μ is in one such given interval."

Confidence Interval of Mean with σ known (Definition)

A **point estimate** is a single number estimate of a population parameter:
sample mean $\bar{x}$ estimates population mean μ.

Confidence Interval of Mean with σ known (Definition)

A **point estimate** is a single number estimate of a population parameter:
sample mean $\bar{x}$ estimates population mean μ.

An **interval estimate** is an interval to estimate a population parameter.

Confidence Interval of Mean with σ known (Definition)

A **point estimate** is a single number estimate of a population parameter: sample mean $\bar{x}$ estimates population mean μ.

An **interval estimate** is an interval to estimate a population parameter.

The **level of confidence** c is the probability that the interval estimate contains the population parameter, when we resample many times.

Confidence Interval of Mean with σ known (Definition)

A **point estimate** is a single number estimate of a population parameter:
sample mean $\bar{x}$ estimates population mean μ.

An **interval estimate** is an interval to estimate a population parameter.

The **level of confidence c** is the probability that the interval estimate contains
the population parameter, when we resample many times.

Sampling distribution for $n \geq 30$ is approx. normal so c is the area under the standard normal curve between **critical values** $\pm z_c$**:**

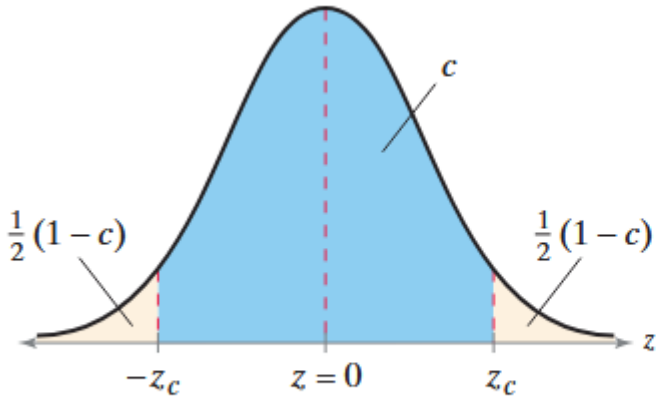
Confidence Interval of Mean with σ known (Definition)

A **point estimate** is a single number estimate of a population parameter: sample mean $\bar{x}$ estimates population mean μ.

An **interval estimate** is an interval to estimate a population parameter.

The **level of confidence c** is the probability that the interval estimate contains the population parameter, when we resample many times.

Sampling distribution for $n \geq 30$ is approx. normal so c is the area under the standard normal curve between **critical values $\pm z_c$:**



If $c = 90\%$:	
$c = 0.90$	Area in blue region
$1 - c = 0.10$	Area in yellow regions
$\frac{1}{2}(1 - c) = 0.05$	Area in one tail
$-z_c = -1.645$	Critical value separating left tail
$z_c = 1.645$	Critical value separating right tail

Confidence Interval of Mean with σ known (Definition)

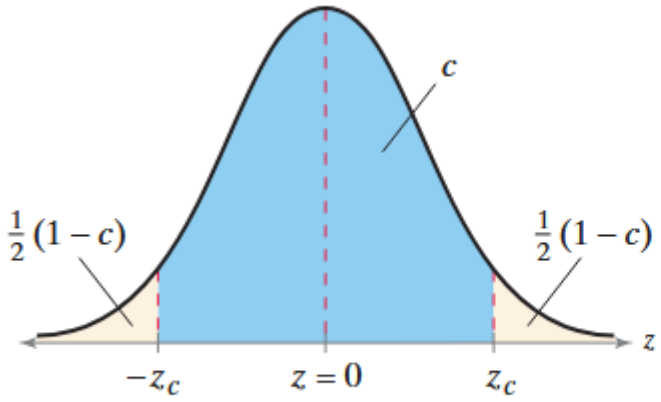
A **point estimate** is a single number estimate of a population parameter: sample mean $\bar{x}$ estimates population mean μ.

An **interval estimate** is an interval to estimate a population parameter.

The **level of confidence c** is the probability that the interval estimate contains the population parameter, when we resample many times.

Sampling distribution for $n \geq 30$ is approx. normal so c is the area under the standard normal curve between **critical values $\pm z_c$:**

$z_c = \text{qnorm}\left(\frac{(1-c)}{2}\right)$ ... command in R



If $c = 90\%$:	
$c = 0.90$	Area in blue region
$1 - c = 0.10$	Area in yellow regions
$\frac{1}{2}(1 - c) = 0.05$	Area in one tail
$-z_c = -1.645$	Critical value separating left tail
$z_c = 1.645$	Critical value separating right tail

Confidence Interval of Mean with σ known (Definition)

A **point estimate** is a single number estimate of a population parameter: sample mean $\bar{x}$ estimates population mean μ.

An **interval estimate** is an interval to estimate a population parameter.

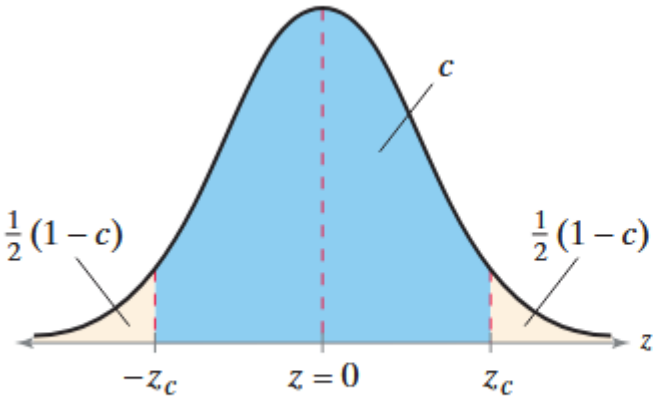
The **level of confidence c** is the probability that the interval estimate contains the population parameter, when we resample many times.

Sampling distribution for $n \geq 30$ is approx. normal so c is the area under the standard normal curve between **critical values $\pm z_c$:**

$z_c = \text{qnorm}\left(\frac{(1-c)}{2}\right)$ ... command in R

Common critical values:

Level of Confidence	z_c
90%	1.645
95%	1.96 ≈ 2
99%	2.575
68%	≈ 1



If $c = 90\%$:	
$c = 0.90$	Area in blue region
$1 - c = 0.10$	Area in yellow regions
$\frac{1}{2}(1 - c) = 0.05$	Area in one tail
$-z_c = -1.645$	Critical value separating left tail
$z_c = 1.645$	Critical value separating right tail

Confidence Interval of Mean with σ known (Facts)

The **c-confidence interval for population mean μ** obtained from an SRS with sample mean $\bar{x}$ is

$$\bar{x} - E < \mu < \bar{x} + E$$

with **margin of error**

$$E = z_c \, \sigma_{\bar{x}} = z_c \frac{\sigma}{\sqrt{n}}$$

when the following conditions are met:

- **10% Condition:** $N \geq 10n$.
- **Approx. normal:** Pop. distribution is normal or $n \geq 30$.

Confidence Interval of Mean with σ known (Facts)

The **c-confidence interval for population mean μ** obtained from an SRS with sample mean $\bar{x}$ is

$$\bar{x} - E < \mu < \bar{x} + E$$

with **margin of error**

$$E = z_c \, \sigma_{\bar{x}} = z_c \frac{\sigma}{\sqrt{n}}$$

when the following conditions are met:

- **10% Condition:** $N \geq 10n$.
- **Approx. normal:** Pop. distribution is normal or $n \geq 30$.

"We are c% confident that μ is in this interval",

" This is the c% confidence interval for μ"

Confidence Interval of Mean with σ known (Example 1)

The **c-confidence interval for population mean μ** obtained from an SRS with sample mean $\bar{x}$ is

$$\bar{x} - E < \mu < \bar{x} + E$$

with **margin of error**

$$E = z_c \, \sigma_{\bar{x}} = z_c \frac{\sigma}{\sqrt{n}}$$

when the following conditions are met:

- **10% Condition:** $N \geq 10n$.
- **Approx. normal:** Pop. distribution is normal or $n \geq 30$.

"We are c% confident that μ is in this interval",

" This is the c% confidence interval for μ"

Example 1.
Construct a 95% confidence interval for mean age of all Rowan students given pop. is normal, $\sigma = 2$ years, and an SRS of size $n = 4$ with $\bar{x} = 21$ years.

Confidence Interval of Mean with σ known (Example 2)

The $\boldsymbol{c}$**-confidence interval for population mean** $\boldsymbol{\mu}$ obtained from an SRS with sample mean $\bar{x}$ is

$$\bar{x} - E < \mu < \bar{x} + E$$

with **margin of error**

$$E = z_c\, \sigma_{\bar{x}} = z_c \frac{\sigma}{\sqrt{n}}$$

when the following conditions are met:

- **10% Condition:** $N \geq 10n$.
- **Approx. normal:** Pop. distribution is normal or $n \geq 30$.

"We are *c*% confident that μ is in this interval",

" This is the *c*% confidence interval for μ"

Example 2. (Minimal sample size)
Rowan student pop. is normal and $\sigma = 2$ years. Find minimal sample size to be 95% confident the sample mean is within 0.5 year of pop. mean.

Confidence Interval (Interpretation)

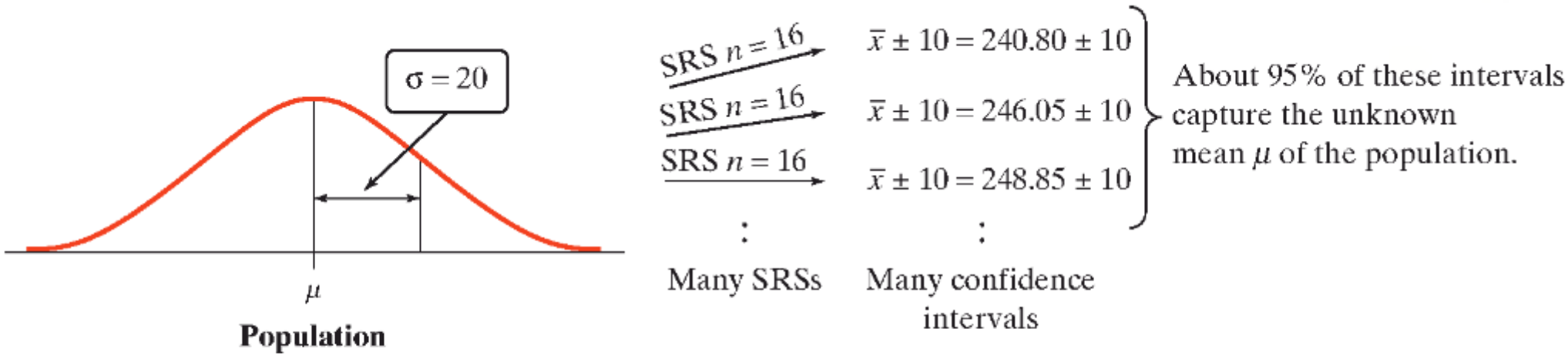


FIGURE 8.4 To say that $\bar{x} \pm 10$ is a 95% confidence interval for the population mean μ is to say that, in repeated samples, about 95% of these intervals capture μ.

Confidence Interval (Interpretation)

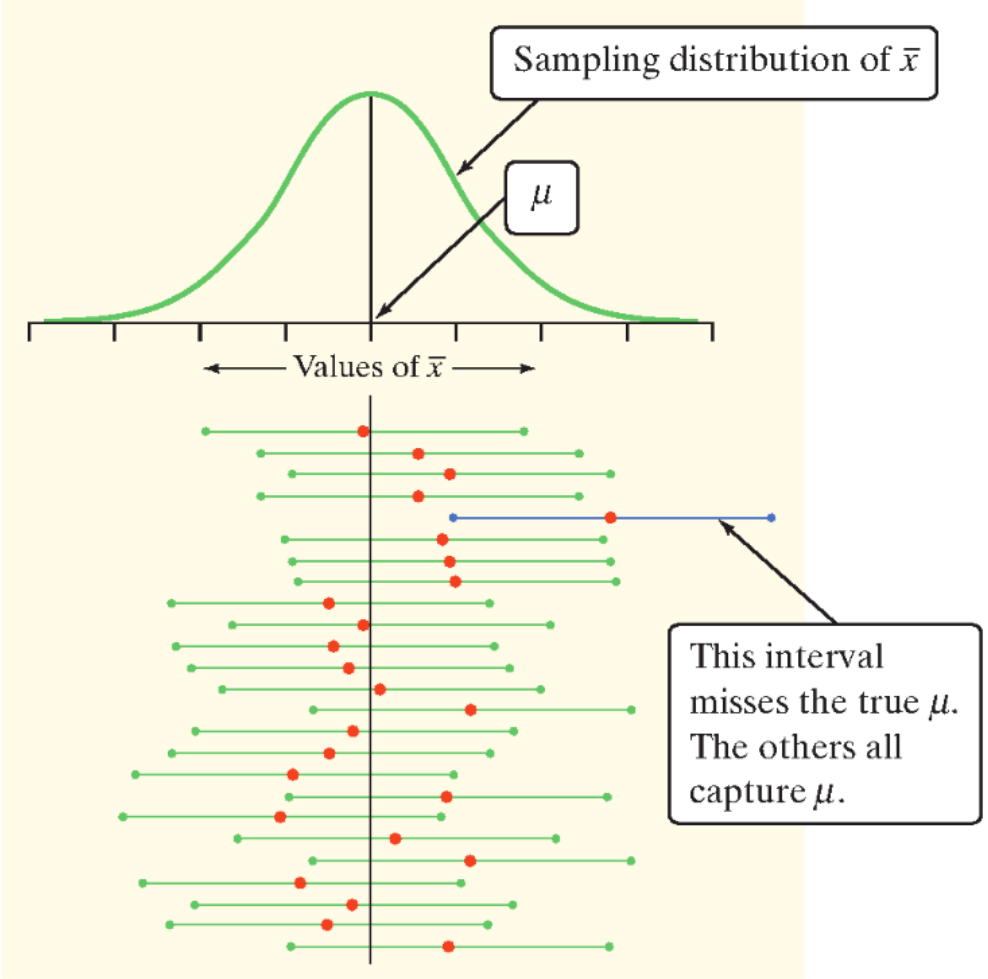


FIGURE 8.5 Twenty-five samples of the same size from the same population gave these 95% confidence intervals. In the long run, about 95% of samples give an interval that captures the population mean μ.

Being *95% confident* is shorthand for:

"If we take many samples of the same size from this population, about 95% of them will result in an interval that captures the true parameter value μ."

(For pop. proportion)

Reference: [ES] 6.3

Confidence Interval of Proportion (Facts)

The $\boldsymbol{c}$**-confidence interval for pop. proportion** $\boldsymbol{p}$ obtained from an SRS with sample proportion $\hat{p}$ is

$$\hat{p} - E < p < \hat{p} + E$$

with **margin of error**

$$E = z_c \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

when the following conditions are met:

- **10% Condition:** $N \geq 10n$.
- **Large counts:** $n\hat{p} \geq 10$ and $n(1-\hat{p}) \geq 10$

Confidence Interval of Proportion (Facts)

The $\boldsymbol{c}$**-confidence interval for pop. proportion** $\boldsymbol{p}$ obtained from an SRS with sample proportion $\hat{p}$ is

$$\hat{p} - E < p < \hat{p} + E$$

with **margin of error**

$$E = z_c \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

when the following conditions are met:

- **10% Condition:** $N \geq 10n$.
- **Large counts:** $n\hat{p} \geq 10$ and $n(1-\hat{p}) \geq 10$

Example 1.
A survey of 1550 U.S. adults finds 1054 use FB. Find a 95% confidence interval for the population proportion.

Confidence Interval of Proportion (Facts)

The $\boldsymbol{c}$**-confidence interval for pop. proportion** $\boldsymbol{p}$ obtained from an SRS with sample proportion $\hat{p}$ is

$$\hat{p} - E < p < \hat{p} + E$$

with **margin of error**

$$E = z_c \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

when the following conditions are met:

- **10% Condition:** $N \geq 10n$.
- **Large counts:** $n\hat{p} \geq 10$ and $n(1-\hat{p}) \geq 10$

To find **minimal sample size** n to ensure $E <$ some number, solve for n. If $\hat{p}$ is unknown, use $\hat{p} = 0.5$.

Confidence Interval of Proportion (Facts)

The $\boldsymbol{c}$**-confidence interval for pop. proportion** $\boldsymbol{p}$ obtained from an SRS with sample proportion $\hat{p}$ is

$$\hat{p} - E < p < \hat{p} + E$$

with **margin of error**

$$E = z_c \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

when the following conditions are met:

- **10% Condition:** $N \geq 10n$.
- **Large counts:** $n\hat{p} \geq 10$ and $n(1-\hat{p}) \geq 10$

To find **minimal sample size** n to ensure $E <$ some number, solve for n. If $\hat{p}$ is unknown, use $\hat{p} = 0.5$.

Example 2. (Minimum sample size)
Find min. sample size to be 95% confident the sample proportion is within 2% of true value.